ENGO 531/632

Advanced Photogrammetric and Ranging Techniques

Laboratory Assignment 4

Close-range Photogrammetric Measurement

Date given: 23 October 2026

Date due: 5 November 2026 at 23.59
Late group submissions will not be accepted;
Late individual submissions lose 10% per day

Marks: This assignment constitutes 15% of your final grade.

Objective

The objective of this lab is to perform a simple measurement project with the mobile phone camera you calibrated in Lab 3. You will select an object or scene to measure, design a network, collect images, extract measurements and assess the accuracy of your work.

For students in ENGO 531

This assignment is to be performed by the group formed for Labs 1-3.

For students in ENGO 632

This is an individual assignment.

Assignment

Data for this assignment are not provided. You will collect data with your own camera that you have already calibrated. You will use the pre-determined calibration parameters from Lab 3 as constants in this assignment.

Select an object (or a scene) somewhere within the Engineering complex to measure. This could be a part of a structure, a façade or a sculpture. To keep the project tractable, the object should not be too large or complex. Objects/scenes must be approved by the TA. You should not need more than eight images to achieve complete coverage of the object/scene. You should also not require more than twenty object points.

Design a network to measure your selected object. You should avoid the placement of artificial targets on the object itself, so choose an object with many natural features that are distinct from their surroundings such as corners. Keep in mind, however, that you will need to measure distances between some of your chosen features. Define a coordinate system and determine approximate coordinates for your feature points. This can be done by total station or even tape measure. The choice is yours.

Collect images of your target field with your camera in the same capture mode as you used for Lab 3. Be sure to carefully determine the approximate EOP values of each image. This can be done in the same way as in Lab 3. Make field notes and sketches as you did for Lab 3.

Measure at least five (5) "important" dimensions of your object of interest using a tape measure. These might be the length, width and depth of a bookshelf, for example. These will serve as independent reference distances for accuracy assessment. You will also measure the corresponding feature points defining these dimensions in the imagery to determine their coordinates. Compute the distances from the photogrammetrically determined coordinates.

In this project, you will define the datum as was done in part B of Lab 2. Use parameter observations to tightly constrain all six of the EOPs for one of the images in your network. Measure an additional (i.e., sixth) distance between two object points and include it as an observation in your bundle adjustment for scale definition. This distance should be different from the five reference lengths.

Identify and measure the points in the imagery as you did in Lab 3. Perform any necessary transformations of the image coordinates. Before you proceed to the bundle adjustment, it is strongly recommended that you overlay your measured image points with IDs on each image to ensure that the target labelling is correct.

Run two bundle adjustments with IOPs fixed as constants:

1. The first is with the nominal IOPs that you used as approximate values for Lab 3.
2. The second is with the IOPs you determined in Lab 3.

As with Lab 3, you should also perform outlier detection and removal.

Analyze the results of each adjustment. Compare the dimensions you independently measured by tape and those determined photogrammetrically. Perform the comparison using results from both sets of IOPs.

Submissions

You are to report on the findings of your object measurement by addressing the following aspects:

- Description of the object you measured
- Geometric network design
- Bundle adjustment results and analyses
    - Overall summary of the network
    - Network quality when the calibrated IOPs are used
    - Any differences in the adjustment results due to use of the nominal IOPs
    - Check distance results and analyses for the two sets of IOPs
- What conclusion can be drawn from the check distance results?

Note that the quantities you analyze are for you to decide, but your analyses are expected to be detailed and justified. It is strongly suggested that you include plots and tables to support your observations. Submissions should include field notes and a sketch of the data capture

Submissions are to be uploaded as a single PDF file to the D2L Dropbox by the specified date and time. For ENGO 531 students, only one group member needs to make a submission but be sure to indicate all members of the group on the title page.

END OF LAB 4